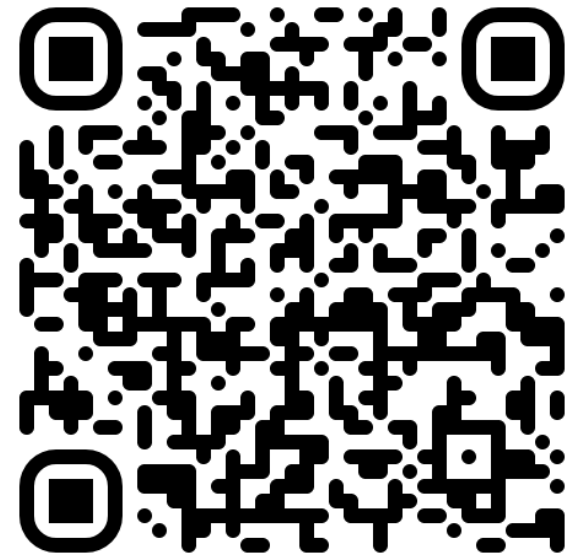


Facilitating Broad Propagation of Computation Integration into the Undergraduate Physics Curriculum using Institutional Change Models

On Behalf of the Computation Integration Collaboration

Eric Hudson

The Pennsylvania State University



Computation is a standard way of doing physics

Theory, Experiment, Computation

- Model • Simulate • Analyze • Visualize • Predict
- Integration = not confined to a programming course
- Why? Need time



A nine-institution collaboration for sustainable computation integration



- 20 collaborators • 30 ambassadors (15 unassigned)
- Together we are developing tools to help departments integrate computation into their curriculum
 - I'll discuss three



We are building a framework for computational learning in physics



Annika Peter



Antonio Boveia
OSU

- Pre-award work over past year
- Surveys → Candidate Competencies → Consensus

Competency 2 – tier consensus (auto-populated from each institution's "Your tier call")

Each cell mirrors that institution's tier call, live. Tallies fold "Agree" into the listed tier, so endorsements and explicit matches count together. "Consensus?" flags rows where collaborators differ

Sub-area	Skill	Listed tier	Penn State	Ohio State	Bridgewater	UMN	UW-Stout	Concordia	St. Thomas	IU
2a. Statistics and Data Analysis										
	Mean, variance, and standard deviation of a dataset	Core	Agree	Agree	Agree	Agree	Agree	Agree	Agree	Agree
	Linear least-squares fitting; interpret parameters and their uncertainties	Core	Agree	Agree	Agree	Agree	Agree	Agree	Agree	Agree
	Chi-squared for a data-model comparison; interpret χ^2 and the p-value	Core	Agree	Agree	Agree	Agree	Agree	Agree	Agree	Optional
	Generate and interpret probability distributions (Gaussian, Poisson, uniform)	Core	Optional	Agree	Agree	Agree	Agree	Agree	Optional	Agree
	Non-linear least-squares fitting	Optional	Core	Agree	Agree	Core	Agree	Agree	Core	Agree
	Covariance and correlation	Optional								

We will produce instruments to assess student proficiency



Diana Sachmpazidi
RIT

- Ground competencies in course artifacts
 - Artifact collection both internal & external (PICUP, ...)
- Build construct maps and prototype items
- Refine through interviews and expert review
- Pilot, analyze & validate

We will produce case studies & broadly applicable CI change models



Charles Henderson
NCSU

- Change Model: Kotter's 8-Step Model (1996)
 - Develop a Sense of Urgency, Build a Coalition ...
 - Local Documentation – Quarterly Status Reports
 - Interviews and Shared Discussions
 - Case Study Development
- Practical “How-to” Contextualized Guidance

We will use a propagation change model to facilitate CI spread

3 Phase Propagation Change Model:

- Interactive Development
 - 9 Member Institutions (2.5% of US BA/BS)
 - 30 Ambassadors
- Interactive Dissemination
 - Papers, Conferences, Workshops
 - PICUP, PhysPort, EP3, ...
 - 30 Ambassadors
- Supporting Adopters



Andy Gavrin
IU Indianapolis



Shaul Hanany
U Minnesota



Marie Lopez del Puerto
U St Thomas

15/30 Ambassador Slots Remain



- Quarterly Virtual Meetings (~2 hrs)
 - Project Updates
 - Discussions
- Disseminate in Your Department

Thanks for
your
attention

Interested? Please Contact Us
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